

Krylov Residual Certificates for Directional Stein Tails

A First Arnoldi Replacement for the Norm Envelope

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Abstract

RH-61 showed that the finite phase-aware tail of a physical packet can be orders of magnitude smaller than the geometric envelope obtained from the largest normalized operator norm. This paper introduces the first directional replacement: an Arnoldi residual certificate for the power $S^L z$ entering a Stein tail. If

$$AV = VH + ge_m^*, \quad z = V\beta,$$

then we prove the exact identity

$$A^L z = VH^L \beta + \sum_{j=0}^{L-1} A^{L-1-j} g e_m^* H^j \beta.$$

Propagating the single Arnoldi defect with an ordinary norm gives a rigorous upper depending on the actual Krylov coefficients. It is exact at Arnoldi breakdown and extends columnwise to Hilbert–Schmidt sources.

Three deterministic models test the mechanism. In a slow/fast surrogate calibrated to the RH-61 endpoint, the ordinary norm envelope is about 2000 times the exact power at $L = 32$, while a one-vector Krylov certificate is essentially exact. In a nonnormal four-step chain, the one-vector gain improves from about 220 to 84, and full Krylov termination is exact. The RH-60 two-block model exposes the remaining problem: a crude one-vector residual propagation can be worse than the ordinary envelope. Krylov structure is therefore a viable route, but the residual propagator must itself become directional or weighted. No production physical-family theorem, Stage A1 closure, self-adjoint operator, arithmetic trace formula, or Hilbert–Polya conclusion is claimed.

Keywords: Arnoldi method; Krylov residual; Stein tail; directional Hardy energy; nonnormal matrix; small noise.

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1 Introduction

The current small-noise spectral route requires directional Hilbert–Schmidt Hardy bounds rather than a global contraction estimate [3]. RH-60 retained a finite phase Gram and completed its tail with a positive Stein metric. RH-61 then showed that the metric’s ordinary norm envelope can be extremely pessimistic even when the actual directional tail is tiny.

The natural next object is the Krylov range generated by the actual tail source. Arnoldi methods provide a finite-dimensional relation between the full operator and its projected Hessian matrix [1, 2]. The relation is especially useful here because the source is fixed and the question is a power applied to that source, not a worst-case resolvent norm.

This paper is deliberately a first layer. We prove the algebraic residual identity, formulate a computable upper, and test it on small models. The propagation of the Arnoldi defect is still bounded by an ordinary operator norm. A failure of that final step is recorded as a route boundary rather than hidden by increasing the Krylov dimension.

Contributions and boundary

1. We prove the Arnoldi power identity in a form directly usable inside a Stein tail.
2. We derive a positive directional power and Stein-tail certificate.
3. We prove exact termination at Krylov breakdown and a columnwise Hilbert–Schmidt extension.
4. We audit slow/fast, two-block, and nonnormal-chain models and isolate the residual-propagation bottleneck.

All production-looking numbers in this paper are finite-dimensional binary64 model computations. The RH-61-calibrated model matches a stored contraction gap only; it is not the folded-Gaussian physical operator.

2 Arnoldi power identity

Let $A \in \mathbb{C}^{n \times n}$ and let $z \neq 0$. Run modified Gram–Schmidt Arnoldi from $v_1 = z/\|z\|$ for k steps. Write $V = [v_1, \dots, v_k]$, let $H = V^*AV$ be the projected Hessenberg matrix, and let g be the final Arnoldi residual. Thus

$$AV = VH + ge_k^*, \quad z = V\beta, \quad \beta = \|z\| e_1. \quad (1)$$

At exact or certified breakdown, $g = 0$. Otherwise g is the unnormalized next Arnoldi vector multiplied by its subdiagonal coefficient.

Theorem 2.1 (Arnoldi power identity). *Under (1), for every integer $L \geq 0$,*

$$A^L z = VH^L \beta + \sum_{j=0}^{L-1} A^{L-1-j} g e_k^* H^j \beta. \quad (2)$$

Proof. For every vector u ,

$$AVu = VHu + ge_k^* u.$$

Apply this relation successively to $u = H^j \beta$ and telescope:

$$A^{j+1}V\beta = A^jVH\beta + A^jge_k^*\beta.$$

Summing the resulting one-step identities from $j = 0$ to $L - 1$ gives (2). The case $L = 0$ is immediate. $\square$

Proposition 2.2 (Directional power certificate). *Let $\rho \geq \|A\|$. Then*

$$\|A^L z\| \leq \|VH^L \beta\| + \|g\| \sum_{j=0}^{L-1} \rho^{L-1-j} |e_k^* H^j \beta|. \quad (3)$$

If $g = 0$, the right side equals $\|H^L \beta\|$ and the identity is exact.

Proof. Apply the triangle inequality to (2) and use $\|A^u g\| \leq \rho^u \|g\|$ for each $u \geq 0$. $\square$

The bound is directional in two distinct ways. The projected power follows the actual starting vector through H , and the residual contribution is weighted by the scalar coefficients $e_k^* H^j \beta$. A common replacement of all coefficients by $\|H\|^j \|\beta\|$ would erase the first advantage.

3 Stein-tail form and extensions

Suppose a normalized packet tail has the form

$$t_L = \sqrt{\kappa} \|S^L z\|_{\mathfrak{S}_2}, \quad \kappa \geq 0. \quad (4)$$

Applying Proposition 2.2 to each source column and summing the resulting bounds gives a valid Hilbert–Schmidt tail certificate.

Corollary 3.1 (Krylov Stein tail). *Let C_ℓ be the power certificate in Proposition 2.2 for a column z_ℓ of a source matrix Z . Then*

$$\sqrt{\kappa} \|S^L Z\|_{\mathfrak{S}_2} \leq \sqrt{\kappa} \sum_{\ell} C_\ell. \quad (5)$$

If the Arnoldi process for every column breaks down, the right side is exact.

Proof. The Hilbert–Schmidt norm is bounded by the sum of the column Euclidean norms. Apply Proposition 2.2 column by column. Breakdown makes each residual term vanish. $\square$

The column sum is conservative. A block Arnoldi implementation can retain cross-column Gram information and replace it by a Frobenius residual. That is a later layer; the present theorem keeps the certificate scalar and transparent.

Remark 3.2 (Boundary). *The residual term in (3) still uses $\rho^{L-1-j} \|g\|$. If ρ is close to one, or exceeds one because of nonnormal transient growth, the residual certificate may remain large. Thus Arnoldi projection alone is not a physical-family tail theorem. It exposes exactly which factor must be replaced next.*

4 Finite-model audit

The pilot uses three deterministic models. The first is a diagonal slow/fast surrogate with slow contraction 0.993861320134583, matching the left RH-61 endpoint gap, and a source whose slow component is only 5×10^{-4} . The second is the two-scalar-block model used by the RH-60 Arb audit. The third is an upper-triangular four-step chain designed to expose nonnormal transient growth.

For each model we compare the ordinary upper $\|A\|^L \|z\|$ with the Krylov bound at $L = 32$. The full Krylov dimension is allowed to terminate exactly.

Table 1: Power-norm upper gains at $L = 32$.

model	$\ A\ $	ordinary gain	Krylov $k = 1$ gain	full Krylov gain
RH-61 slow/fast surrogate	0.993861	2000.00	1.000000	1.000000
RH-60 two-block model	0.765889	15.25	66.41	1.000000
four-step nonnormal chain	1.110729	219.82	84.05	1.000000

The first row demonstrates the desired mechanism: the ordinary norm sees a slow mode with tiny source amplitude, while the Krylov residual records that directional fact. The second row is a genuine warning. With only one Arnoldi vector, the residual is propagated using a norm bound and the result is worse than the baseline. The third row shows partial improvement even when the ordinary norm exceeds one; spectral stability alone is not enough to make the residual propagator useful.

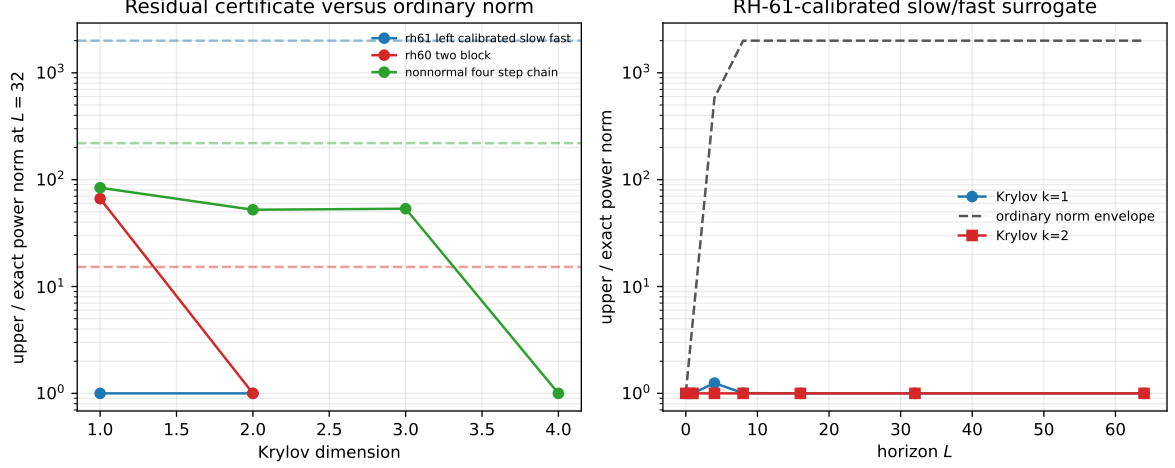


Figure 1: Krylov residual audit. Left: upper-to-exact gains as the Krylov dimension grows; dashed lines are ordinary norm gains. Right: the RH-61-calibrated slow/fast surrogate across horizons. Full Krylov termination is exact, while the one-vector certificate tracks the directional power closely in this model.

A 256-bit Arb calculation verifies the Arnoldi identity and the one-step upper on the RH-60 two-block model. It does not interval-validate any folded-Gaussian production matrix.

5 Route consequence

RH-62 establishes an intermediate certificate:

$$\text{physical source} \longrightarrow \text{Krylov projection} \longrightarrow \text{Arnoldi residual} \longrightarrow \text{Stein tail.} \quad (6)$$

It removes the need to use the largest norm on the projected component. However, the remaining residual propagation is still a worst-case step.

The next minimal theorem is therefore not simply to take a larger Krylov space. It is to replace $\rho^{L-1-j} \|g\|$ by a second directional certificate for $A^u g$. Two natural choices are a nested Arnoldi residual recursion and an observability-weighted residual Gramian. Either one must preserve a positive upper and admit an a posteriori stopping test.

1. A nested residual certificate should be tested first on the RH-60 two-block model, where the one-step bound fails.
2. A block version should then retain cross-column phase information rather than summing columns absolutely.
3. Only after those finite-dimensional tests pass should the construction be transferred to the physical folded-Gaussian family.

6 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no von Mangoldt or prime-power trace formula, and no completed-zeta spectral identity. It makes no Hilbert–Polya or Riemann-hypothesis claim. The TPC branch is not used.

7 Reproducibility

The directory contains the Arnoldi algebra, tests, three-model pilot, 256-bit Arb audit, figures, hashes, and publication artifacts. The main commands are shown below.

```
pytest -q -p no:cacheprovider
python experiments/run_krylov_tail_pilot.py
python experiments/run_arb_krylov_audit.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The finite-dimensional identity and certificate are analytic. The model tables are deterministic floating-point evidence, and the Arb scope is the displayed two-block example only. Stage A1, Stage A4, a physical-family directional tail theorem, a self-adjoint Hilbert–Polya operator, a prime-power trace formula, and a zeta-zero identity remain open.

References

- [1] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4 edition, 2013.
- [2] Yousef Saad. *Iterative Methods for Sparse Linear Systems*. Society for Industrial and Applied Mathematics, Philadelphia, 2 edition, 2003.
- [3] Liang Wang. Two-pole Hilbert–Schmidt hardy energies for small-noise range resolvents. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-50-two-pole-hilbert-schmidt-hardy.